Solvency Field Theory
V11 Derivation Companion - Part 1

Foundations, Compton Maintenance, the Mesoscale Bridge, and the Modified Friedmann Equation

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Companion draft prepared from V11 Master Part 1 and the historical V10 derivation archive

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Abstract

This companion volume supports Sections 1–6 of the V11 Master Paper. The master paper is written to say what Solvency Field Theory is claiming; this companion is written to show the mathematical spine underneath those claims. It fixes notation, separates the established physical inputs from the SFT closure axiom, derives the Compton maintenance count unit, formalizes the mesoscale bridge from discrete ticking to the continuum operation density, and derives the modified Friedmann ratio

$$\frac{H_{\mathrm{SFT}}^2}{H_{\mathrm{std}}^2} = \frac{1}{4\pi^2}.$$

The emphasis is not rhetorical. Each load-bearing statement is expressed as a claim, an assumption set, a derivation block, and a closure status. Where V11 Master gives the face of the theory, this document gives the first layer of the bones.

Contents

1 Role of this companion **2**
- 1.1 Source hierarchy for Part 1 . . . 2
- 1.2 What this part proves . . . 2
- 1.3 Status language . . . 2

2 Notation and conventions **3**

3 Claim ledger for Master Sections 1–6 **4**

4 Axioms and the master identity **5**
- 4.1 The three inherited inputs . . . 5
- 4.2 The SFT closure axiom . . . 5
- 4.3 Relation to the Hubble-time count equation . . . 5

5 Compton Maintenance Postulate **7**
- 5.1 Unitary phase versus irreversible bookkeeping . . . 7
- 5.2 Massive particles and photons as write patterns . . . 7

6 One-particle weak-field lag profile **8**

7 Two particles and N particles: additivity and gravity as gradient **9**

8 The continuum source: pressure as transport burden **10**
- 8.1 Particle ensemble in a coarse-graining volume . . . 10
- 8.2 Dust and radiation limits . . . 10

9 Covariant operation density **11**

10 Gravity as local gradient; expansion as global sum **12**

11 Modified Friedmann equation from the Hubble count **13**

12 Where the two 2π factors enter **15**

13 Five formulations, one Friedmann normalization **16**

14 Open items carried forward from Part 1 **17**

15 Part 1 summary **17**

1 Role of this companion

The V11 Master Paper is organized in the order of physical inevitability: one particle, two particles, many particles, continuum, gravity as a gradient, expansion as a sum. This companion mirrors that structure, but with the derivational burden restored.

1.1 Source hierarchy for Part 1

The following hierarchy is used whenever older notes and the V11 Master use different language.

1. **V11 Master Part 1 Final** is the current claim and status source.
2. **V10.10 Mesoscale Bridge** is the current bottom-up source for the discrete-to-continuum construction.
3. **V10 Companion Combined Reading Volume** is the historical full derivation archive for the count equation, modified Friedmann ratio, and supporting identities.
4. **V7.1/V7.4 Compton and count notes** are provenance sources used only where they are not superseded by V11.

1.2 What this part proves

Companion Part 1 supports the following master-paper spine:

$$\nu_C = \frac{mc^2}{h_{\mathrm{P}}}, \tag{1}$$

$$\rho_{\mathrm{ops}} = \frac{T_{\mu\nu}u^\mu u^\nu + h^{\mu\nu}T_{\mu\nu}}{h_{\mathrm{P}}} = \frac{\rho + 3p}{h_{\mathrm{P}}}, \tag{2}$$

$$H = \frac{\dot{S}}{I}, \tag{3}$$

$$\frac{H^2_{\mathrm{SFT}}}{H^2_{\mathrm{std}}} = \frac{1}{4\pi^2}. \tag{4}$$

1.3 Status language

This companion uses the same status language as the V11 Master.

Status	Meaning
Exact	Follows algebraically or variationally inside the stated assumptions.
Derived/recovered	Reached through the stated derivation, with any branch or unit assumptions identified.
Structurally identified	The correct mathematical object has been isolated, but full background-independent or fully rigorous closure remains open.
Effective closure	The result is supported by a controlled effective model or robustness basin, while first-principles histories remain open.
Research target	The result is not yet carried to the floor and is explicitly left open.

2 Notation and conventions

The companion must avoid a specific ambiguity: in the master paper, the letter h appears both as Planck's constant and as the spatial projector $h_{\mu\nu}$. In this companion, Planck's constant is always written as h_{P}.

$$h_{\mathrm{P}} \equiv \text{Planck's constant}, \tag{5}$$

$$\hbar = \frac{h_{\mathrm{P}}}{2\pi}, \tag{6}$$

$$\ell_{\mathrm{P}}^2 = \frac{\hbar G}{c^3} = \frac{h_{\mathrm{P}} G}{2\pi c^3}. \tag{7}$$

The metric signature is taken to be $(-,+,+,+)$. A unit timelike clock congruence satisfies

$$u^\mu u_\mu = -1. \tag{8}$$

The spatial projector orthogonal to u^μ is

$$h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu, \qquad h^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu. \tag{9}$$

For a perfect fluid in the clock frame,

$$T_{\mu\nu} = (\rho + p) u_\mu u_\nu + p g_{\mu\nu}, \tag{10}$$

where ρ is energy density and p is pressure. When cosmological equations are written in standard SI form, ρ denotes energy density. When the reduced convention $8\pi G/(3c^2) = 1$ is used, the same symbol denotes the reduced Friedmann source.

For a Hubble sphere,

$$R_H = \frac{c}{H}, \tag{11}$$

$$V_H = \frac{4\pi}{3} R_H^3 = \frac{4\pi c^3}{3H^3}, \tag{12}$$

$$A_H = 4\pi R_H^2 = \frac{4\pi c^2}{H^2}, \tag{13}$$

$$I_H = \frac{A_H}{4\ell_{\mathrm{P}}^2} = \frac{\pi c^2}{H^2 \ell_{\mathrm{P}}^2}. \tag{14}$$

3 Claim ledger for Master Sections 1–6

Master	Load-bearing claim	Companion proof block	Status
1–3	The first three inputs are inherited physics; the fourth is the SFT closure axiom.	Sec. 4.1–4.2	Structural scaffold
3	$H = \dot{S}/I$ is the SFT master identity inside the stated scaffold.	Sec. 4.3	Closure identity
4	A massive rest particle supplies a natural count rate $\nu_C = mc^2/h_{\mathrm{P}}$.	Sec. 5.1	Exact count unit
4	Irreversibility attaches to boundary bookkeeping, not to unitary bulk phase evolution alone.	Sec. 5.2	Clarification
5.1	A point maintenance source gives a $1/r$ weak-field lag profile.	Sec. 6	Weak-field recovery
5.2–5.3	Many lag fields add in the weak-field limit, and their gradient gives gravity.	Sec. 7	Linearized limit
5.3–5.4	Pressure is the continuum transport burden of moving states.	Sec. 8	Derived projection
5.4	The continuum operation density is $(\rho + 3p)/h_{\mathrm{P}}$.	Sec. 8–9	Derived
5.5	Gravity is the local gradient and expansion is the global sum.	Sec. 10	Structural identification
6	The Hubble count equation gives $H^2_{\mathrm{SFT}}/H^2_{\mathrm{std}} = 1/(4\pi^2)$.	Sec. 11–12	Derived
6.3	Five formulations recover the same Friedmann normalization.	Sec. 13	Consistency ledger

4 Axioms and the master identity

Master claim. The master equation $H = \dot{S}/I$ is not a standard theorem of existing thermodynamics. It is the native SFT closure identity once irreversible information production, finite holographic capacity, and geometric accommodation are taken as the governing scaffold.

4.1 The three inherited inputs

The foundation begins with three imported physical inputs.

1. **Landauer cost.** Information erasure or irreversible information processing has a thermodynamic cost.
2. **Thermal time.** Macroscopic time is tied to irreversible state change rather than an externally imposed absolute clock.
3. **Holographic capacity.** The maximum information content of a region is bounded by boundary area in Planck units:
$$I_{\max} = \frac{A}{4\ell_{\mathrm{P}}^2}. \tag{15}$$

These inputs alone do not imply SFT. The fourth axiom is the SFT closure step.

4.2 The SFT closure axiom

The SFT closure axiom states that the universe accommodates the irreversible maintenance ledger geometrically. In cosmology, that accommodation is expansion. Thus the relevant closure is not simply "information exists" but

$$\text{irreversible production of new ledger demand} \quad \longleftrightarrow \quad \text{geometric growth of available capacity.} \tag{16}$$

Let $\dot{S}$ be the total information-production rate measured in SFT count units, and let I be the available holographic capacity of the active cosmological boundary. The dimensionless occupancy pressure per unit time is $\dot{S}/I$. The SFT closure axiom identifies this with the cosmological expansion rate:

$$\boxed{H = \frac{\dot{S}}{I}.} \tag{17}$$

Status. Equation (17) is the SFT master identity inside the stated scaffold. It is not presented as a model-independent theorem of standard thermodynamics. It is the closure law that defines SFT's information-maintenance interpretation of expansion.

4.3 Relation to the Hubble-time count equation

Multiplying Eq. (17) by one Hubble time $t_H = H^{-1}$ gives

$$\dot{S} t_H = I. \tag{18}$$

Writing the total processed information over one Hubble time as $S(t_H)$ yields the count equation

$$\boxed{S(t_H) = I(t_H).} \tag{19}$$

This is the settlement form of the master identity: over the natural cosmological time window, the ledger production equals the boundary capacity.

5 Compton Maintenance Postulate

Master claim. A massive rest particle maintains its quantum state at the Compton frequency, and this frequency supplies the primitive count rate for SFT maintenance.

For a particle of rest mass m, the rest energy is

$$E_0 = mc^2. \tag{20}$$

The ordinary frequency associated with this energy is

$$\nu_C = \frac{E_0}{h_{\mathrm{P}}} = \frac{mc^2}{h_{\mathrm{P}}}. \tag{21}$$

Thus the SFT one-particle rest maintenance rate is

$$\boxed{\dot{s}_1 = \nu_C = \frac{mc^2}{h_{\mathrm{P}}}.} \tag{22}$$

5.1 Unitary phase versus irreversible bookkeeping

In ordinary quantum mechanics, phase evolution is unitary. A state factor $\exp(-iEt/\hbar)$ evolves reversibly under the Schrödinger equation. SFT does not claim that this bulk phase rotation is itself thermodynamically irreversible.

The irreversibility is assigned to the boundary registration required to maintain persistent physical state on a finite holographic ledger in an expanding universe. The Compton cycle supplies the natural count unit; the ledger update supplies the thermodynamic cost.

The distinction can be summarized as follows:

$$\text{bulk phase evolution}: \quad \text{unitary, reversible,} \tag{23}$$

$$\text{holographic ledger registration}: \quad \text{irreversible, thermodynamic,} \tag{24}$$

$$\text{SFT count unit}: \quad \text{one complete Compton cycle.} \tag{25}$$

Status. This is a conceptual closure statement. The companion treats the Compton rate as the count unit and assigns thermodynamic cost to holographic bookkeeping, not to unitary phase evolution in isolation.

5.2 Massive particles and photons as write patterns

The universal statement behind the postulate is not restricted to rest mass. Any physical state of energy E carries a natural operation rate

$$\dot{s}_E = \frac{E}{h_{\mathrm{P}}}. \tag{26}$$

For a rest particle, $E = mc^2$ and the write pattern is localized same-cell refresh. For a photon, $E = h_{\mathrm{P}}\nu$ and the write pattern is spatial handoff. The distinction is therefore not a different fee, but a different way of spending the same c-limited processing budget.

6 One-particle weak-field lag profile

Master claim. In the weak-field limit, a point maintenance source produces the standard $1/r$ potential profile. The master paper states this narratively; the companion identifies the Green-function step.

Let a point particle at the origin have maintenance rate

$$\dot{s}_1 = \frac{mc^2}{h_{\mathrm{P}}}. \tag{27}$$

In the weak-field, linearized limit, the local lag response is modeled by a Poisson-type equation for a scalar lag potential ψ:

$$\nabla^2\psi = \mathcal{K}\,\dot{s}_1\,\delta^{(3)}(\mathbf{x}), \tag{28}$$

where $\mathcal{K}$ is the conversion constant between operation rate and geometric lag. The Green function of the three-dimensional Laplacian satisfies

$$\nabla^2\left(-\frac{1}{4\pi r}\right) = \delta^{(3)}(\mathbf{x}). \tag{29}$$

Therefore Eq. (28) gives

$$\psi(r) = -\frac{\mathcal{K}\dot{s}_1}{4\pi r}. \tag{30}$$

The Newtonian potential of a point mass is

$$\Phi_N(r) = -\frac{Gm}{r}. \tag{31}$$

Matching the weak-field lag gradient to the Newtonian acceleration fixes the conversion constant in the weak-field limit. Thus the $1/r$ profile is not assumed directly; it is the Green-function solution of the linearized local lag equation for a point maintenance source.

Result. A point Compton maintenance source yields a $1/r$ weak-field lag potential, with the normalization fixed by the Newtonian limit.

Status. Weak-field recovery. The companion does not claim to derive G from first principles in this block; it shows how the SFT maintenance source maps to the standard point-source Green function once the weak-field operator and Newtonian normalization are imposed.

7 Two particles and N particles: additivity and gravity as gradient

Master claim. In the weak-field limit, information costs add, lag profiles superpose, and gravity is the gradient of the summed lag field.

For two particles with lag fields ψ_1 and ψ_2, linearity of the weak-field Poisson operator gives

$$\boxed{\psi_{\text{total}}(\mathbf{x}) = \psi_1(\mathbf{x}) + \psi_2(\mathbf{x}).} \tag{32}$$

The acceleration experienced by particle 1 is proportional to the gradient of the total lag field at its location. Its own spherically symmetric self-field contributes no net self-gradient at its center, so

$$\nabla\psi_{\text{total}}\big|_{\mathbf{x}_1} = \nabla\psi_2\big|_{\mathbf{x}_1}. \tag{33}$$

For N particles,

$$\boxed{\psi_{\text{total}}(\mathbf{x}) = \sum_{i=1}^{N} \psi_i(\mathbf{x}),} \tag{34}$$

so the gravitational field at particle j is

$$\boxed{\mathbf{g}_j = -\nabla\psi_{\text{total}}\big|_{\mathbf{x}_j} = -\sum_{i\neq j} \nabla\psi_i\big|_{\mathbf{x}_j}.} \tag{35}$$

For point sources with $\psi_i \propto -m_i/r_i$, Eq. (35) gives the inverse-square law.

Result. In the linear weak-field limit, Newtonian gravity is the gradient of the summed maintenance-debt profile.

Status. This block establishes the mesoscale bridge in the weak-field/superposition regime. Nonlinear strong-field behavior is handled by the exponential composition principle in later companion parts.

8 The continuum source: pressure as transport burden

Master claim. The continuum source $\rho_{\rm ops} = (\rho + 3p)/h_{\rm P}$ is the coarse-grained operation rate of an ensemble of particles whose c-budget is split between same-cell refresh and spatial transport.

8.1 Particle ensemble in a coarse-graining volume

Consider a coarse-graining volume V containing particles with rest masses m_i, Lorentz factors γ_i, and speeds v_i. The energy density is

$$\rho = \frac{1}{V} \sum_{i \in V} \gamma_i m_i c^2. \tag{36}$$

For an isotropic ensemble, the pressure is the average momentum flux:

$$p = \frac{1}{3V} \sum_{i \in V} \gamma_i m_i v_i^2. \tag{37}$$

The factor 1/3 is the angular average over three spatial directions.

SFT assigns two operation-load pieces to the ensemble:

1. occupancy load, the energy cost of state service;
2. transport burden, the momentum-flux cost of handing state between spatial cells.

Thus the coarse-grained operation density is

$$\boxed{\rho_{\rm ops} = \frac{1}{V h_{\rm P}} \sum_{i \in V} \left(\gamma_i m_i c^2 + \gamma_i m_i v_i^2 \right) = \frac{\rho + 3p}{h_{\rm P}}.} \tag{38}$$

The first term in the sum gives $\rho/h_{\rm P}$; the second gives $3p/h_{\rm P}$ by Eq. (37).

8.2 Dust and radiation limits

For pressureless dust, $p = 0$, so

$$\boxed{\rho_{\rm ops}^{\rm dust} = \frac{\rho}{h_{\rm P}}.} \tag{39}$$

For radiation, $p = \rho/3$, so

$$\boxed{\rho_{\rm ops}^{\rm rad} = \frac{2\rho}{h_{\rm P}}.} \tag{40}$$

The factor of two is not a separate rule for photons. It is the equal contribution of occupancy load and isotropic transport burden for a relativistic fluid.

Status. The particle-sum derivation is a mesoscale bridge. A fully rigorous convergence theorem from discrete Compton ticking to the continuum source is listed as a V11 research target.

9 Covariant operation density

Master claim. The particle-sum source has a coordinate-free expression in the local clock frame: $\rho_{\rm ops} = (T_{\mu\nu}u^\mu u^\nu + h^{\mu\nu}T_{\mu\nu})/h_{\rm P}$.

Let u^μ be the clock-frame four-velocity and

$$h^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu \tag{41}$$

be the spatial projector. Define

$$\boxed{\rho_{\rm ops} = \frac{T_{\mu\nu}u^\mu u^\nu + h^{\mu\nu}T_{\mu\nu}}{h_{\rm P}}.} \tag{42}$$

For a perfect fluid in the clock frame,

$$T_{\mu\nu} = (\rho + p)u_\mu u_\nu + p g_{\mu\nu}. \tag{43}$$

The occupancy projection is

$$\begin{aligned} T_{\mu\nu}u^\mu u^\nu &= (\rho + p)(u_\mu u^\mu)(u_\nu u^\nu) + p g_{\mu\nu}u^\mu u^\nu && (44)\\ &= (\rho + p)(1) + p(-1) && (45)\\ &= \rho. && (46) \end{aligned}$$

The spatial trace is

$$\begin{aligned} h^{\mu\nu}T_{\mu\nu} &= (g^{\mu\nu} + u^\mu u^\nu)T_{\mu\nu} && (47)\\ &= T + T_{\mu\nu}u^\mu u^\nu. && (48) \end{aligned}$$

For metric signature $(-,+,+,+)$,

$$T = g^{\mu\nu}T_{\mu\nu} = -\rho + 3p. \tag{49}$$

Therefore

$$h^{\mu\nu}T_{\mu\nu} = (-\rho + 3p) + \rho = 3p. \tag{50}$$

Substituting into Eq. (42) gives

$$\boxed{\rho_{\rm ops} = \frac{\rho + 3p}{h_{\rm P}}.} \tag{51}$$

Result. The covariant source is exactly the continuum form of occupancy load plus transport burden.

Status. Source content derived in the clock frame. The full action-level variation of the implicit χ-dependence in u^μ and $J^\mu_{\rm ops}$ is carried in Companion Part 2.

10 Gravity as local gradient; expansion as global sum

Master claim. The same operation density appears locally as the gradient of lag and globally as boundary growth.

The local weak-field picture is governed by a differential response operator:

$$\nabla^2\psi \sim \text{local maintenance source.} \tag{52}$$

The gravitational field is the local slope of the lag field,

$$\boxed{\mathbf{g} = -\nabla\psi.} \tag{53}$$

The cosmological picture integrates the same operation density over the Hubble volume:

$$\boxed{\dot{S} = \int_{V_H} \rho_{\text{ops}}\, dV.} \tag{54}$$

The same maintenance activity therefore has two complementary macroscopic readings:

$$\text{local gradient} \longrightarrow \text{gravity,} \tag{55}$$

$$\text{global integral} \longrightarrow \text{expansion through } H = \dot{S}/I. \tag{56}$$

This is the precise mathematical version of the master-paper sentence: gravity is what happens when ticks concentrate; expansion is what happens when one sums them all.

Status. Structural identification. The weak-field gradient statement is supported by the Poisson/Green-function block. The global sum statement becomes exact in the Hubble-count derivation of the next section.

11 Modified Friedmann equation from the Hubble count

Master claim. In homogeneous FLRW dust, the Compton maintenance rate inside the Hubble volume divided by the holographic boundary capacity gives the modified Friedmann ratio $H_{\mathrm{SFT}}^2/H_{\mathrm{std}}^2 = 1/(4\pi^2)$.

For dust, $p = 0$, so

$$\rho_{\mathrm{ops}} = \frac{\rho}{h_{\mathrm{P}}}, \tag{57}$$

where ρ is energy density. The total information-production rate inside the Hubble sphere is

$$\dot{S} = \int_{V_H} \frac{\rho}{h_{\mathrm{P}}}\, dV \tag{58}$$

$$= \frac{\rho}{h_{\mathrm{P}}} \frac{4\pi c^3}{3H^3}. \tag{59}$$

The holographic capacity of the Hubble boundary is

$$I_H = \frac{A_H}{4\ell_{\mathrm{P}}^2} \tag{60}$$

$$= \frac{1}{4\ell_{\mathrm{P}}^2} \frac{4\pi c^2}{H^2} \tag{61}$$

$$= \frac{\pi c^2}{H^2 \ell_{\mathrm{P}}^2}. \tag{62}$$

Insert Eqs. (59) and (62) into the master identity:

$$H = \frac{\dot{S}}{I_H} \tag{63}$$

$$= \frac{\rho}{h_{\mathrm{P}}} \frac{4\pi c^3}{3H^3} \frac{H^2 \ell_{\mathrm{P}}^2}{\pi c^2} \tag{64}$$

$$= \frac{4\rho c \ell_{\mathrm{P}}^2}{3 h_{\mathrm{P}} H}. \tag{65}$$

Multiplying by H gives

$$H^2 = \frac{4\rho c \ell_{\mathrm{P}}^2}{3 h_{\mathrm{P}}}. \tag{66}$$

Using

$$\ell_{\mathrm{P}}^2 = \frac{h_{\mathrm{P}} G}{2\pi c^3} \tag{67}$$

then gives

$$\boxed{H_{\mathrm{SFT}}^2 = \frac{2G\rho}{3\pi c^2}.} \tag{68}$$

The standard flat FLRW dust equation written for energy density ρ is

$$H_{\mathrm{std}}^2 = \frac{8\pi G}{3c^2}\rho. \tag{69}$$

Therefore

$$\boxed{\frac{H_{\mathrm{SFT}}^2}{H_{\mathrm{std}}^2} = \frac{\frac{2G\rho}{3\pi c^2}}{\frac{8\pi G\rho}{3c^2}} = \frac{1}{4\pi^2}.} \tag{70}$$

In reduced Friedmann units where $8\pi G/(3c^2) = 1$, this is written as

$$\boxed{H^2 = \frac{\rho}{4\pi^2}.} \tag{71}$$

Result. The modified Friedmann coefficient is not fitted. It is recovered by combining the Compton maintenance count, the Hubble volume, the holographic boundary capacity, and the Planck-area definition.

Status. Derived for the homogeneous dust branch. Radiation and action-level recovery are treated in the formulation ledger below and in Companion Part 2.

12 Where the two 2π factors enter

The ratio in Eq. (70) can be understood as a product of two independent conversion factors.

The first conversion is the phase-counting factor. SFT counts complete cycles, using h_{P}, while the Planck area uses $\hbar$:

$$h_{\mathrm{P}} = 2\pi\hbar. \tag{72}$$

Thus the Compton count and the Planck-area capacity are not expressed in the same angular unit without a 2π conversion.

The second conversion is the horizon-thermodynamic factor that appears when quantum phase accounting is mapped into gravitational response through horizon thermality. In the V11 master this is carried by the KMS/Unruh structure and is developed in the $4\pi^2$ spine. In this Part 1 derivation it is already visible through the same Planck-area normalization and Hubble-horizon capacity that convert the count into a Friedmann response.

The compact statement is

$$\boxed{4\pi^2 = (2\pi)_{\mathrm{phase}}(2\pi)_{\mathrm{thermal}}.} \tag{73}$$

Status. Part 1 derives the Friedmann ratio explicitly. The full theorem-level decomposition of the two factors is carried in Companion Part 2, mirroring Master Section 8.

13 Five formulations, one Friedmann normalization

Master claim. The V11 Master lists five routes to the same Friedmann normalization. The companion separates exact derivations from branch-level recoveries.

Formulation	Core equation	Status in companion
Master identity	$H = \dot{S}/I$	Closure identity of SFT scaffold.
Hubble count equation	$S(t_H) = I(t_H)$	Algebraically equivalent to the master identity over one Hubble time.
Direct Friedmann ratio	$H^2_{\mathrm{SFT}}/H^2_{\mathrm{std}} = 1/(4\pi^2)$	Derived in Sec. 11 for homogeneous dust.
Constrained minisuperspace action	$C = 0$ imposes count normalization	Historical V10 derivation; summarized in Companion Part 2 where the action is introduced.
Repaired covariant action	$C_{\min} = 0$ with $\beta = 1/(4\pi^2)$	Dust exact; radiation asymptotically exact; full action derivation in Companion Part 2.

The important point is that the same coefficient appears even when the mathematical route changes: direct count law, minisuperspace constraint, and repaired covariant action all preserve the $1/(4\pi^2)$ normalization in the appropriate branch.

Status. Consistency ledger. The routes are not all independent empirical confirmations; they are independent internal formulations recovering the same coefficient within SFT.

14 Open items carried forward from Part 1

Part 1 does not close every mathematical question raised by the master paper. It establishes the first derivation layer and identifies what must be carried into later companion volumes.

Open or deferred item	Where it is handled
Formal convergence of discrete Compton ticking to the continuum operation density	Listed as a research target; Part 1 gives the kinetic/coarse-grained bridge.
Full action variation of J^{μ}_{ops} and hidden χ dependence	Companion Part 2, Lagrangian derivation.
Full $4\pi^2$ theorem-level decomposition	Companion Part 2, structural identity section.
Radiation branch recovery from the repaired action	Companion Part 2, action background reductions.
CMB transfer normalization	Companion Part 3, early-universe sector.
Strong-field exponential composition and repaired-action exterior collapse	Companion Part 3, strong-field sector.

15 Part 1 summary

The Part 1 companion establishes the following derivational chain:

1. The SFT master equation $H = \dot{S}/I$ is the closure identity connecting irreversible information production to geometric boundary accommodation.
2. The Compton frequency mc^2/h_{P} supplies the one-particle maintenance count rate.
3. The irreversibility attaches to holographic ledger registration, not to unitary phase rotation alone.
4. A point maintenance source produces a $1/r$ weak-field lag profile through the Green function of the Poisson operator.
5. Weak-field lag profiles add, and their gradient gives Newtonian gravity.
6. Coarse-graining an ensemble of particles gives the covariant operation density
$$\rho_{\text{ops}} = \frac{T_{\mu\nu}u^{\mu}u^{\nu} + h^{\mu\nu}T_{\mu\nu}}{h_{\text{P}}} = \frac{\rho + 3p}{h_{\text{P}}}.$$
7. Integrating this source over the Hubble volume and dividing by holographic boundary capacity yields
$$\frac{H^2_{\text{SFT}}}{H^2_{\text{std}}} = \frac{1}{4\pi^2}.$$

Part 1 bottom line.

The connecting rod between the Compton Maintenance Postulate and the modified Friedmann equation is counting: one-particle maintenance rates coarse-grain into the operation density, local gradients give gravity, and the global Hubble-volume sum gives expansion.